

A finite noise window for concentration of a shared flow

Deterministic disturbance and steps toward selection

David Galbraith

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Abstract

Can unequal sharing of a driven flow contribute to a physical account of selection? For a bounded reflected diffusion in which channels share a conserved total, we derive the exact joint stationary law and distinguish the population of channel states from the flow they carry. With flow weight g^m and diffusion factor g^q , the population fraction above the pooled half-flow threshold tends to zero as both state range and population grow precisely when $1 \leq q \leq m+1$, for exact powers. At fixed population N it instead approaches $1/(2N)$ inside the window. The upper edge is the established weighted-moment threshold, made exact here for the coupled model. Independent simulations check the stationary predictions on bounded grids. We distinguish this sorting result from spatial channel formation, persistent structures, reproduction and portable heredity. Existing and constructive models identify possible steps between them; they do not yet establish the whole sequence in one physical system. A conditional branching threshold states what further measurements would be needed to predict lineage establishment. Empirical applicability remains untested.

1 The physical question and the upper boundary

The motivating proposal is conditional: disturbance associated with a driven flow could help produce concentrated sharing only within a suitable scaling window, and that concentration could become relevant to selection if states persist and are copied. The window is essential to this proposal. A mechanism that concentrates under every disturbance rule would not establish it.

The research programme asks whether this sequence can be constructed without adding random forcing as a fundamental ingredient: deterministic fine-scale dynamics produce effective variation; variation affects the division of flow; and inherited states use some of that flow to reproduce. This paper separates an exact stationary result from computational thought experiments that explore those connections. The proposed synthesis is not equivalent to the known power-law calculation on which one step depends.

A collection of paths can be counted in two ways: by the number of paths in each state, or by the fraction of throughput they carry. A rare large state can matter little to the first count and greatly to the second. The question here is when that difference becomes extreme as the accessible state range grows.

Noise affects this comparison through residence times. In a scalar driftless Itô diffusion with reflecting boundaries, the stationary density is inversely proportional to diffusivity. State-dependent noise and occupation probabilities have a long history [1, 2]. Weighting a power-law distribution by a power of its argument introduces a standard moment threshold [3]. Neither ingredient is new.

Newman explicitly discusses both the general moment condition and the fraction of a weighted total held in a population tail (Secs. III.B and III.D of Ref. [3]). Their use for a fixed shared flow requires care: each channel's share depends on every other channel, and

the same normalization can change the physical clock of the state dynamics. We give an exactly solvable coupled model in which this dependence is explicit. The result identifies the finite-population correction and makes the observable behind the noise window precise.

For $m = 1$, the upper edge is $q = 2$. Beyond it, increasing the upper state cutoff does not make a vanishing population fraction account for half the pooled flow. This does not mean that flow becomes equally distributed. A small but nonzero fraction can still carry half. Nor is q a noise amplitude: it describes how diffusivity changes with state. Multiplying every diffusivity by one constant changes relaxation times, but leaves all stationary results below unchanged.

2 A model with conservation imposed at every instant

Let $N \geq 1$ channels have dimensionless states $g_i \in [1, R]$, where $R > 1$. The model specifies possible flow routes and their response law; it does not prescribe which routes become dominant. The lattice experiment in Sec. 6 tests differentiation from identical coarse-route states. Spatial formation and connectivity of routes in a material are abstracted out: no spatial energy-gradient field or changing network topology is modeled here. For $m > 0$, define the total weight S and the actual flow through channel i by

$$S(\mathbf{g}) = \sum_{j=1}^N g_j^m, \quad J_i(\mathbf{g}) = I \frac{g_i^m}{S(\mathbf{g})}. \quad (1)$$

Thus $\sum_i J_i = I$ for every configuration, not only in expectation. We choose the interior dynamics

$$dg_i = \sqrt{2D_i(\mathbf{g})} dW_i, \quad D_i(\mathbf{g}) = \frac{N g_i^q}{S(\mathbf{g})}, \quad (2)$$

with independent Wiener processes, normal reflection at both endpoints, and zero interior drift. Boundary local-time terms are implicit. A positive common prefactor has been absorbed into the time unit. The lower cutoff sets the state unit; all powers refer to this specified coordinate.

The normalization in Eq. (2) is a constitutive assumption. On the line $q = m$, $D_i = (N/I)J_i$, so noise variance rate is proportional to actual flow. Away from that line the model keeps the common factor $1/S$. The apparently similar choice $D_i \propto J_i^{q/m}$ instead contains $S^{-q/m}$ and defines different dynamics. Conservation of flow alone does not select either rule.

The Itô convention would follow from an appropriate small-kick limit of conditionally unbiased additive events in g , evaluated at the pre-event state. Such a limit also needs control of temporal correlations and higher moments. Unresolved deterministic dynamics do not automatically satisfy these conditions. Under a nonlinear change $y = h(g)$, Itô's formula adds the drift $D(g)h''(g)$ even when the drift in g is zero. The zero-drift assumption must therefore be justified in the measured coordinate, rather than restored after arbitrary relabelling.

3 Exact stationary population and flow distributions

On $[1, R]$, write

$$Z_R = \int_1^R g^{-q} dg, \quad p_0(g) = \frac{g^{-q}}{Z_R}, \quad \mu_R = \int_1^R g^m p_0(g) dg, \quad f_R(g) = \frac{g^m p_0(g)}{\mu_R}. \quad (3)$$

All quantities are finite at finite R , for every real q .

Proposition 1. *The unique stationary density of Eq. (2) on the reflecting box is*

$$\pi_{N,R}(\mathbf{g}) = \frac{S(\mathbf{g})}{N\mu_R} \prod_{j=1}^N p_0(g_j). \quad (4)$$

Its tagged-channel marginal and its pooled flow density are, respectively,

$$p_{N,R}(g) = \left(1 - \frac{1}{N}\right) p_0(g) + \frac{1}{N} f_R(g), \quad (5)$$

$$\mathbb{E}_\pi \left[\sum_{i=1}^N \frac{J_i}{I} \delta(g - g_i) \right] = f_R(g). \quad (6)$$

Proof. Integration of $S \prod_j p_0(g_j)$ gives $N\mu_R$, establishing normalization. For each coordinate,

$$D_i \pi_{N,R} = \frac{1}{Z_R \mu_R} \prod_{j \neq i} p_0(g_j)$$

is independent of g_i . Every probability current $-\partial_i(D_i \pi_{N,R})$ therefore vanishes, including at the boundaries. The diffusion is uniformly nondegenerate on the connected compact box, giving uniqueness. Expanding the sum S and integrating out the other coordinates gives Eq. (5). Multiplication by $J_i/I = g_i^m/S$ cancels S in the joint law; integration and summation give Eq. (6). $\square$

The calculation also has a simple clock interpretation. If $L_0 = \sum_i g_i^q \partial_i^2$ is the generator of independent reflected channels, the coupled generator is $L = (N/S)L_0$. Configurations with larger S are traversed more slowly relative to that independent process; their stationary weight acquires the factor S . This explains both the solvability and the dependence of the result on the chosen normalization.

An exact independent sampling construction is useful: choose one channel uniformly, draw its state from f_R , and draw the other $N - 1$ states independently from p_0 . Averaging over the chosen channel reproduces Eq. (4). That designation is resampled for each configuration; it does not identify a permanent winner.

The correction matters even in simple cases. For $N = 1$ and $q = m$, the channel carries all the flow and has constant diffusivity. Its stationary density is uniform, not g^{-m} . At fixed R , the tagged marginal approaches p_0 as $N \rightarrow \infty$. The independent-channel calculation is recovered in that limit.

4 The concentration observable and its window

Let a_R be the state threshold above which half of the pooled flow lies:

$$\int_{a_R}^R f_R(g) dg = \frac{1}{2}. \quad (7)$$

Define $P_0(R) = \int_{a_R}^R p_0(g) dg$ and $P_N(R) = \int_{a_R}^R p_{N,R}(g) dg$. Equation (5) gives

$$\boxed{P_N(R) = \left(1 - \frac{1}{N}\right) P_0(R) + \frac{1}{2N}.} \quad (8)$$

This counts a population fraction above one threshold fixed across stationary observations. It is a pooled observable, with each observation weighted equally in physical time.

Theorem 1. For Eqs. (1)–(2) with exact powers and $m > 0$, $P_N(R) \rightarrow 0$ as both $R \rightarrow \infty$ and $N \rightarrow \infty$ if and only if

$$\boxed{1 \leq q \leq m + 1.} \quad (9)$$

At fixed N , the limit inside this window is $1/(2N)$.

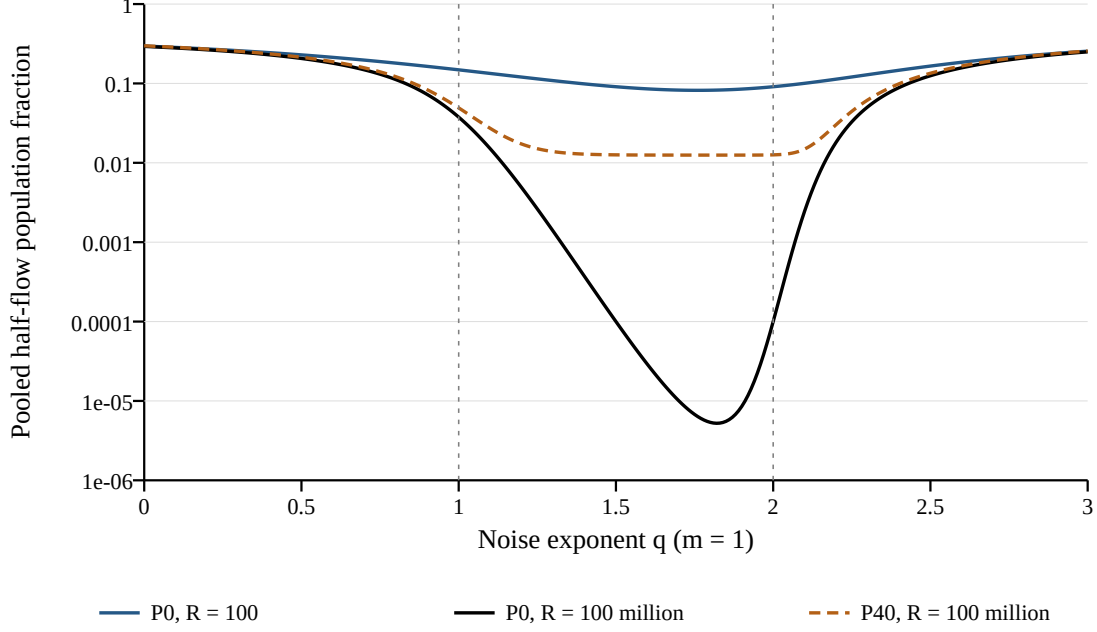


Figure 1: Exact stationary predictions, not empirical measurements or large-range dynamic simulations. The asymptotic window is between the dotted lines. Finite range smooths both boundaries; finite population supplies a floor. P_0 is the large-population limit at each fixed R . The curve at $R = 10^8$ is not the infinite-range limit.

Proof. Put $A = m - q + 1$ and $B = 1 - q$. Direct integration gives

$$a_R = \begin{cases} [(R^A + 1)/2]^{1/A}, & A \neq 0, \\ \sqrt{R}, & A = 0, \end{cases} \quad P_0(R) = \begin{cases} (R^B - a_R^B)/(R^B - 1), & B \neq 0, \\ \log(R/a_R)/\log R, & B = 0. \end{cases} \quad (10)$$

For $q < 1$, $a_R/R \rightarrow 2^{-1/A}$ and $P_0 \rightarrow 1 - 2^{-B/A} > 0$. At $q = 1$, $P_0 \sim \log 2/(m \log R) \rightarrow 0$. For $1 < q < m + 1$, a_R grows proportionally to R , while the normalized population tail above it vanishes. At $q = m + 1$, $P_0 = 1/(R^{m/2} + 1) \rightarrow 0$. For $q > m + 1$, $a_R \rightarrow 2^{1/(q-m-1)}$, so $P_0 \rightarrow 2^{-(q-1)/(q-m-1)} > 0$. Equation (8) supplies the finite- N result and the joint limit. $\square$

Either iterated limit gives this classification, as does any sequence with both R and N tending to infinity. Stationarity is taken at each bounded pair (N, R) first. No unbounded diffusion or uniform equilibration time is asserted. The same classification holds for any fixed pooled flow fraction strictly between zero and one, with the corresponding fraction divided by N replacing $1/(2N)$.

The integrals behind the two edges are $\int g^{-q} dg$ and $\int g^{m-q} dg$. At the lower edge the population normalization is logarithmic; at the upper edge the flow normalization is logarithmic. The upper-edge population fraction itself decays as a power of R . Figure 1 displays the finite-range expressions.

4.1 Pooled concentration is not a snapshot winner count

In a single configuration, rank shares in descending order and count the smallest number of whole channels whose shares sum to at least one half. Call that count divided by N H_N . It always satisfies $H_N \geq 1/N$. In contrast, P_N can approach $1/(2N)$ because its fixed threshold need not carry half the flow in every snapshot. These observables must not be substituted

for one another. The numerical checks measure $\mathbb{E}H_N$ separately against the exact joint sampler. Neither statistic measures how long a particular channel remains dominant. Stationary concentration alone establishes neither persistent structure nor Darwinian selection.

For example, at $m = 1$, $q = 2$, $R = 10^8$, $P_0 \simeq 0.000100$, whereas $P_{40} \simeq 0.012598$. This is a substantial relative correction to the tail fraction, even though $1/N$ is small.

4.2 Limiting slopes do not determine the endpoints

The inclusive endpoints require exact powers or additional tail assumptions. On $g \geq e$, consider the unnormalized flow weight $w(g) = g$ and noise factor $d(g) = g^2(\log g)^2$. Their limiting log-log slopes are $m = 1$ and $q = 2$, but $\int_e^\infty w(g)/d(g)dg$ converges. The upper-edge separation fails. This example also fits the coupled construction with $S = \sum_i w(g_i)$ and $D_i = Nd(g_i)/S$: replace p_0 by the normalized $1/d$ and f_R by the normalized w/d in the proof. Logarithmic corrections do not change the strict interior power inequalities, but can decide the boundary cases. An estimated slope near an edge is consequently insufficient to classify a measured system.

5 What would make the result physically useful?

The upper bound excludes a proposed mechanism when the measured disturbance scaling is too steep relative to the flow weight, provided the rest of the model applies. It also distinguishes state-dependent disturbance from monotonic reinforcement intuition. At $m = 1$, the large-range, large-population half-flow fraction vanishes for $q = 2$, but approaches $1/8$ for $q = 2.5$. Above the boundary it can still be very small, especially close to $q = 2$. The mathematical transition is the loss of a vanishing fraction, not an abrupt return to equal sharing in a finite apparatus.

The basic inequality is a standard power-law moment condition [3]. Its value here is the connection to actual conserved shares and the observable correction in Eq. (8). These make a conditional prediction that can be checked. They do not establish a new general law of nature, and no priority claim for the elementary threshold is made.

One possible microscopic route to state-dependent variance uses contact patches. If each patch contributes an unbiased additive perturbation in the same measured coordinate, n independent patches give variance proportional to n ; perfectly coherent patches within a channel give variance proportional to n^2 . Inferring $q = m$ or $q = 2m$ additionally requires $n \propto g^m$, fixed coupling per patch, and a suitable time scale. It does not derive the common $1/S$ clock in Eq. (2). In particular, a rule proportional to squared actual flow would have a different normalization. Coherence within a channel must also be distinguished from correlations between channels, which violate the independent-noise assumption.

A fair physical test needs simultaneous channel states and individual flows, with a fixed total and a stable channel population. It must estimate both conditional drift and variance rate in the chosen coordinate, check cross-channel correlations, and allow enough relaxation before comparing stationary statistics. For Eq. (2), conditioning on the configuration or accounting for S is necessary: a regression on g_i alone need not recover q . Drift, measurement noise, jumps, changing populations and limited state range can each prevent the test from addressing this model. Power-law fits themselves also need checks against alternatives [4].

A useful experiment would vary the state dependence of disturbance across the upper edge while measuring the finite-range prediction, rather than expecting an infinite-limit discontinuity. No physical dataset is used as validation in this paper.

6 From shared flow to selection

Concentration is organisation across channels, not necessarily structure within a channel. Fixed pipes can share flow unequally without forming a whirlpool; a whirlpool can persist without a few structures dominating the total flow. The proposed connection therefore contains five distinct questions, which need not occur in a fixed chronological order:

Step	Evidence and remaining distinction
Flow paths form	Erosion models develop channels from a rough slope. Routing and erosion laws are specified; geometry develops through them.
Flow sorts	The theorem gives an exact pooled concentration window for its specified dynamics. It does not establish spatial structure or persistent winners.
A pattern persists	The existing lattice-Boltzmann model examines circulation behind a supplied pole. Persistence of shape is distinct from inheritance of a particular perturbation.
A pattern reproduces	Reaction–diffusion spots can divide through local reaction and transport laws. Division alone does not establish inherited variants.
Information travels	A carrier may recreate a property elsewhere. Existing token models assume its copying and expression; a physical carrier must supply them.

The flow-structure example uses local lattice updates, a maintained inflow and one fixed irregularity: the pole. Directional particle populations stream to neighbours and relax through collisions. Lattice-Boltzmann rules recover Navier–Stokes behaviour in the appropriate hydrodynamic limit [12]; non-binary values alone do not ensure that limit. The pole’s persistence is supplied, while the circulation is produced by the dynamics. The repaired model shows returning tracers in its nearly steady wake, absent in the low-drive, linear and no-pole controls. Numerical details remain in the repository.

The deterministic fine-lattice experiment addresses a source of disturbance. Starting forty coarse routes at identical states, three fine-state seeds give final top-three flow shares of .615, .463 and .529, against .075 initially and in frozen controls. No random numbers are drawn after initialization. A periodic control also concentrates, so chaos is not shown necessary. Deterministic effective noise has established mathematical precedents [5, 6]; this lattice’s drift, correlations and stochastic convention have not been matched to Eq. (2). It consequently does not test the complete two-sided window.

For a constructive step, the capability must follow from the interaction rules, rather than an instruction to copy a detected structure. The repository retains controlled spatial examples and numerical checks separately. Its Gray–Scott example reproduces known spot multiplication [8, 9]; related chemical spot replication has been observed experimentally [10]. These are supplied reaction networks and finite prepared perturbations, not an estimate of spontaneous origin rates. They do not yet connect the concentration window to heredity. Molecular templating, including peptide self-replication [11], provides relevant physical precedent for a carrier simpler than a complete genetic system; it is not a demonstrated product of these flow models.

Once a copying mechanism exists, its consequences have a precise mathematical question. If usable work arrives at rate ηJ and a copy costs E , the ideal long-run copying rate is $\eta \bar{J}/E$, subject to losses. In a rare-lineage branching approximation, let T describe inherited-state changes, d disappearance, and P viable offspring states per copying attempt. Established

N	Grid intervals	Occupancy CDF	Flow CDF	Mean snapshot fraction
8	4	0.017014	0.021730	0.011423
8	8	0.020299	0.024502	0.005979
40	8	0.009496	0.008539	0.001585

Table 1: Largest final absolute errors across the six exponent pairs and four initial conditions for each retained run, all on $[1, 3]$. CDF errors compare with the exact lattice law. The snapshot reference uses 100,000 independent joint-law samples per pair. Seeds are pooled at the final observation time.

branching theory [7] gives the conditional establishment threshold

$$E < \eta\rho[(\text{diag } d - T)^{-1} \text{diag}(J)P]. \quad (11)$$

The repository checks this calculation against independent event simulations. It assumes exponential copying clocks and a fixed background while the lineage is rare. It does not derive T , d and P from the same substrate as the concentration model, or predict finite-population takeover. Establishing that connection, with perturbations becoming inherited variations through the physical dynamics, is the remaining task. Rare origins need not be simulated at historical waiting times, but their possibility requires a physical construction and their spread requires a separate establishment test.

7 Numerical checks of the stationary result

The repository implements a continuous-time nearest-neighbour chain on a uniform state lattice with spacing h . Each allowed jump of channel i by $+h$ or $-h$ has rate

$$\lambda_i(\mathbf{g}) = \frac{Ng_i^q}{S(\mathbf{g})h^2}. \quad (12)$$

Outward boundary jumps are absent. Exponential holding times and rate-weighted event selection simulate this finite chain without a time-step approximation. The lattice invariant law is exactly Eq. (4) with sums replacing integrals. Grid refinement remains necessary for a continuous-state comparison. Observations are taken at fixed physical times, because sampling only jump events would bias residence-time statistics.

Eight automated tests include detailed balance on every edge of an enumerated three-channel, three-state system; independent marginal calculations; the single-channel case; the finite-population quantile identity; sampling checks; continuum integral comparisons; grid refinement; and relaxation from both boundaries in a simple control. The retained ensemble runs use six exponent pairs: $(1, 0.5)$, $(1, 1)$, $(1, 1.5)$, $(1, 2)$, $(1, 2.5)$ and $(2, 4)$. For each pair they use lower-boundary, upper-boundary, log-uniform and exact stationary initial conditions, with three seeds and 128 independent ensembles per seed. Physical observation times are 0, 1, 4, 16.

All retained final errors are below the declared tolerance of 0.05 (Table 1). This is an acceptance criterion, not a confidence interval. Separate exact-law calculations check grid refinement and evaluate large state ranges; they are not dynamical runs at those ranges. The checks support the implementation on the tested bounded grids. They do not establish large-range equilibration times or empirical applicability.

The earlier independent-walker check is excluded from this evidence. It started log-uniformly, which is already the target stationary law at $q = 1$, and did not establish relaxation. It is retained in the repository as a labelled legacy diagnostic. The exact power integrals remain valid.

8 Conclusion

The conserved-flow model gives an exact concentration window and a finite-population correction. The upper bound states when rare large states cease to dominate pooled throughput; its mathematics is established. The physical proposal is the connection from such sorting to structures whose variations persist and reproduce. The repository makes several steps concrete and keeps their assumptions separate. A complete account must derive the connections under consistent physical rules and test them independently; concentration alone is not a derivation of selection.

Code and reproducibility. Manuscript sources, model implementations, tests and retained data are at <https://github.com/divadwg/selection-as-physics>. Build and reproduction commands are in `paper/README.md`; the repository separates current evidence from historical interpretations and secondary models.

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